



The effect of font readability on the Moses illusion: A replication study

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ABSTRACT

This research is a replication study that sought to verify whether the readability of a font has an effect on the Moses illusion detection. It was designed to stimulate information retrieval from memory and confuse retrieval with a text's erroneous wording. Undergraduates aged 19–30 ($N = 87$, 80% women) were presented with two questions, one of which contained distorted information. We assumed that a difficult-to-read font would facilitate error detection, as it increases the focus of attention on the text. However, unlike the original study, we were unable to find support for this hypothesis, as font readability did not significantly affect error detection. In the difficult-to-read condition, 43% of participants reported an error, while, in the easy-to-read condition, errors were detected by 37% of the participants. Unlike the original study, our research results do not support the hypothesis that the visual presentation of a text affects the automatic retrieval of information from memory. This study clarifies the effect of text readability on error detection taking into consideration the role of long-term memory and visual perception.

1. Introduction

This replication study examines the phenomenon in which people do not notice errors in a presented text when the information is processed fluently. Past studies have shown that fluently processed material is perceived as accurate (Schwarz, 2004), is often accepted as true (McGlone & Tofigbaksh, 2000), and does not encourage rigorous critical review (Claypool et al., 2004). According to recent research, the fluency of processing information in a text might be affected by the readability of the font it is presented in (Song & Schwarz, 2008; Eslick, Fazio, & Marsh, 2011) and the font size (Price, McElroy, & Martin, 2015). However, other studies used different strategies to visually modify their texts and failed to find evidence that would support the influence of visual distortion on detecting semantic error in written content (Eslick et al., 2011; Price et al., 2015).

Strategies of processing communicated information are a traditional subject of interest in psychological research (Park & Reder, 2004). The Moses illusion is an example of an uncritical review of presented material. The Moses illusion was observed in the original study of Erickson and Mattson (1981), in which they found that most people answer “two” to the question of “How many animals of

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each kind did Moses take on the Ark?”, despite knowing that Noah is the protagonist of the biblical narrative of Noah’s Ark. The Moses illusion is concerned with what affects the consistency of information processing.

There are various explanations of the Moses illusion in scholarly literature. The theory which has received most support is the two-pass process model, in which the detection of error involves two consecutive steps (Park & Reder, 2004). In the first step, a potential inaccuracy or mismatch of information is detected, while, in the second step, the attention is focused on closer inspection of the text (Park & Reder, 2004; Reder, 1987, 1988). Only if an inaccuracy is detected in the first step will the attention be focused on critical inspection of the text in the second step. If the inaccuracy goes undetected, the error in the text will go unnoticed.

Another explanation lies in the third phase of the RI-Val model (Cook & O’Brien, 2014), which bears similarities to the partial matching process and the role of shallow processing (Cook et al., 2018). Cook et al. (2018) also proposed that shallow processing could be integrated into the RI-Val model as a result of an incomplete validation process. Moreover, Park & Reder (2004) offer a partial matching explanation, which is based on semantic similarity, since conceptual representations of both protagonists (Moses and Noah) share several features that can lead to failure to detect the distortion in the question. Nevertheless, this study focuses more on the replication of an original study (Song & Schwarz, 2008) and testing the influence of text font on the illusion, rather than on general Moses illusion phenomena.

Cognitive processing fluency depends on both the form and content of the received information. As far as content is concerned, thematic overlap seems to play an essential role, as it enables the fluent processing of information that is similar to a well-known topic (Erickson & Mattson, 1981; Park & Reder, 2003). In the case of the Moses illusion, it is the thematic closeness of both the characters (Moses and Noah) in the Old Testament and the general knowledge that their stories are related to water. Thus, the fluency of information processing is influenced by the perceived familiarity of the subject, which is amplified by physical characteristics such as a high figure-ground contrast, long exposure time, and easy-to-read fonts (Kelley & Rhodes, 2002; Reber, Schwarz, & Winkielman, 2004; Schwarz, 2004).

Experiments that test variables affecting the fluency of text processing (Song & Schwarz, 2008; Eslick et al., 2011; Price et al., 2015) confirm the influence of the content and the formal elements of the text (e.g., font readability). The assumption is that material with an easy-to-read text will appear to be familiar (Kelley & Rhodes, 2002; Schwarz, 2004), and thus will often be accepted as true (McGlone & Tofiqbaksh, 2000; Reber & Schwarz, 1999) and not encourage rigorous critical review (Claypool et al., 2004; Garcia-Marques & Mackie, 2001). The combination of familiarity and fluency leads to an increase in erroneous judgment (Whittlesea, Jacoby, & Girard, 1990) and a strong belief in the correctness of the information (Koriat & Levy-Sadot, 2001).

Song and Schwarz (2008) tested the effect of font readability on information processing. They tested the Moses illusion in two conditions. One group was presented with a text written in a difficult-to-read font and the other group with a text written in an easy-to-read font. Their results confirmed the aforementioned assumption, i.e., in the group with the easy-to-read font a lower proportion of participants detected an error than in the group with the difficult-to-read font. Other studies have tested the same theory using a different colour (Eslick et al., 2011) and font size (Price et al., 2015). However, the studies with different colours and font sizes did not support the hypothesis of text processing disruption. They found that neither colour nor font size leads to sufficient activation of cognitive abilities to suppress previously formed associations that are automatically activated when presented with a known topic, which, in this case, was a biblical story (Eslick et al., 2011). This result can be interpreted as in accordance with Reder and Kusbit (1991), who demonstrated that people tolerate discrepancies in a text and adjust the presented information according to a representation of the outside world.

The literature indicates that motivation may play a key role in the Moses illusion tasks when avoiding an automatic response of respondents. Marsh and Fazio (2006) add that to detect an error, one must have sufficient motivation and the ability to disobey automatic processing heuristics. To increase motivation in their experiment, Marsh and Fazio (2006) provided one group of participants with two buttons designated “next” and “error” that they could choose for each question. The participants were asked to report an error whenever the question contained false information. The results revealed that participants who were active in selecting the answer for each question, detected the error more often than participants who were only instructed to detect errors at the beginning of the experiment. This evidence suggests that a warning at the beginning is insufficient motivation to detect the error, but that motivation increases when people label each question. However, a recent study conducted by Speckmann and Unkelbach (2021) points to the fact that providing a financial incentive is not sufficient motivation to detect discrepancies in a text. Therefore, the specific type of motivation to disobey automatic processing heuristics is contextually dependent.

Given the above-mentioned findings, we were doubtful as to whether the decreased readability of the font would disrupt word processing fluency. Moreover, it was unclear whether the font change would provide sufficient motivation for the boost in attention that initiates error detection. This was one reason we decided to replicate the experiment by Song and Schwarz (2008). Another reason was connected with the low power of the original experiment. The original sample size was small ($N = 32$), and the results of the Moses illusion experiment were close to the significance threshold ($z = 2.5$, $p < 0.02$). The authors of the study also pointed out that the mood of the participants could have been a possible intervening variable, but they did not check its influence on the results. Previous research confirms the effect of mood on attention to detail (Schwarz & Clore, 2007). When people are sad, they are more inclined to detail-oriented processing and elaborate on the presented material in more detail (Bless, Bohner, Schwarz, & Strack, 1990). Moreover, they often take an analytical approach to information processing (Alter, Oppenheimer, Epley, & Norwick, 2007).

Therefore, we decided to test again the original hypothesis (H: When a question is written in a difficult-to-read font, it is easier to detect an error in the text, when compared to a question written in an easy-to-read font.) on a larger sample and included the mood of the participants as an intervening variable. We designed the study as a close replication of Song and Schwarz (2008) using the same instructions, the same materials, and same analyses as the original study. Were we to find the same results as in the original study, it would provide stronger evidence for the effect of font readability on the detection of semantic distortion. On the other hand, non-

significant results would mean that the results of the original, frequently cited, study (Song & Schwarz, 2008) should be interpreted with caution.

2. Materials and method

2.1. Pilot testing

As with the original study by Song and Schwarz (2008), we conducted a pilot study to compare the readability of the two fonts. We used the same easy-to-read font as in the original study, Arial, and Allura was chosen as the difficult-to-read font, as it is identical to Brush Script MT, the font used in the original study, apart from the fact that it has Czech diacritics.

The pilot testing included 16 participants (in the original study, there were only 10 participants). To calculate the pilot testing sample size, we derived a 75 % effect size from the original study (Cohen's $d = 2.005$; the original Cohen's $d = 2.673$) with test power of $1 - \beta = 0.8$ and $\alpha = 0.01$. We presented the sentence "Switzerland is famous for cuckoo clocks, banks, and pocket knives" written in Allura (in font size 16) to eight undergraduates selected by convenience sampling. Another eight participants read the same sentence in Arial font (in font size 12). The font size was identical for both fonts. Participants rated the ease with which they were able to read the text on a 7-point scale (1 = very difficult to read, 7 = very easy to read).

Using t-tests in IBM SPSS, the assumption that the Allura font was a more difficult to read than Arial was verified. There was a significant difference between the two fonts, $t(14) = 2.743$, $p = .017$, where Arial was easier to read ($M = 6.125$, $SD = 1.246$) than Allura ($M = 4.625$, $SD = 0.916$).

2.2. Experiment

2.2.1. Sample

The sample consisted of 87 Czech undergraduates from various fields (80 % of whom were women), who were selected by convenience sampling. The age of the participants ranged from 19 to 30 years ($M = 20.73$; $SD = 1.93$).

We used G*Power (G*Power 3.1.9.7, Faul et al., 2007) to calculate the required sample size for a z-test of proportion difference. With a test power of $1 - \beta = 0.9$ and significance level at $\alpha = 0.05$, we wanted to be able to detect an even smaller difference in proportion than was found in the original study. In the original study, 6% of participants in the easy-to-read condition and 40% of participants in the hard-to-read condition detected the error. We set the expected proportions to 6% and 30% (i.e., 75% of the proportion in the hard-to-read condition in the original study), which determined a minimum sample size of 84 participants. We invited a total of 87 participants, in case any participant withdrew from the experiment or chose to withdraw their data.

2.2.2. Procedure

We used the same procedure as that of Song and Schwarz (2008). We started with a questionnaire that collected demographic data (age, gender, and field of study) and included a question to ascertain the participant's current mood, which we measured on a nine-point scale. We then gave the participants instructions and presented them with two printed open-ended trivia questions in Czech. The first of these was a question to which participants could provide a correct answer ("Which country is famous for cuckoo clocks, chocolate, banks, and pocket knives?"), whereas the second question did not have a valid answer ("How many animals of each kind did Moses take on the Ark?" i.e., the Moses illusion). Participants were randomly assigned to one of two experimental conditions (both questions were printed in either an easy-to-read font – Arial, $N_{\text{Arial}} = 43$; or in a difficult-to-read font – Allura, $N_{\text{Allura}} = 44$). There was no time limit for answering the questions. For more information on the procedure, see the Appendix.

Prior to the experiment, the participants were given the following instruction in Czech: "You will read a couple of trivia questions and answer them. You can write the answer in the blank space provided. In case you do not know the answer, please write 'don't know'. You may or may not encounter ill-formed questions which do not have correct answers if taken literally. For instance, you might see the question 'Where does Jan Masaryk come in the order of Czechoslovak presidents?' The fact is that Jan Masaryk was never president, and, therefore, the correct answer to this question is 'I can't answer'. The only difference between our study and the original one was the language of the instructions and the modification of the sample question that served as an example of a question that should be answered 'I can't answer'. We modified the original question on Gerald Ford's presidency to suit the Czech environment.

The procedure was conducted in a university study room. For both the pilot and main study, we acquired informed consent from the participants. We ensured that none of the participants were able to see another person's questionnaire and thus discover the purpose of the experiment. After all participants had completed the questionnaire, we collected them in and debriefed the participants.

2.2.3. Measurement

The dependent variable was the detection of the error in the text. The distortion was identified when a participant answered 'I can't answer' to the Moses illusion question. The independent variable was the text's readability, which was in two conditions; the easy-to-read font (Arial) and the difficult-to-read font (Allura).

3. Results

3.1. Preliminary analysis

Before analysing the main statistics, we tested whether the participants' mood affected the results. The groups were similar in mood ($M_1 = 6.03$; $SD_1 = 1.33$; $M_2 = 5.97$; $SD_2 = 1.4$), $t(62) = 0.183$; $p = .855$. Using logistic regression, we tested whether mood affected the detection of the error in the experimental condition. The results of the logistic regression showed that mood had no significant effect on error detection (see [Tables 1 and 2](#)). We concluded that mood does not seem to have any intervening effects on the study results, and thus did not include it in the further analysis.

3.2. Main analyses

As with the original study, we tested the hypothesis using proportion z-tests. We received 87 answers to both questions.

Regarding the distorted question ("How many animals of each kind did Moses take on the Ark?"), 43% of the participants detected the error in the difficult-to-read font, i.e., they wrote: "I can't answer". When the font was easy-to-read, 37% of the participants detected the error, $z = -0.57$; $p = .569$. In total, 54% of the participants in the easy-to-read condition and 45% of the participants in the difficult-to-read condition answered "two" in response to the Moses question, $z = 0.75$; $p = .453$. The answer "I don't know" was given by 9% of participants with the easy-to-read font and 11% of participants with the difficult-to-read font, $z = -0.32$; $p = .749$. None of these results were significant at a 0.05 alpha level. Thus, we did not find support for the tested hypothesis.

We found differences in some answers to the undistorted question ("Which country is famous for cuckoo clocks, banks, and pocket knives?"). When the font was easy-to-read 93% of participants answered correctly, i.e., Switzerland, while 77% of participants answered correctly when the font was difficult to read, $z = 2.06$; $p = .039$. Furthermore, in the difficult-to-read condition, 9% of participants named a country other than Switzerland, whereas in the easy-to-read condition no one named any other country than Switzerland, $z = -2.02$; $p = .043$.

In the easy-to-read condition 7% of participants reported that they did not know the answer, while, in difficult-to-read condition, 14% of the participants did not know the answer, $z = -1.02$; $p = .308$.

Thus, the Moses illusion did appear in the distorted question, but it did not differ on the basis of the readability of fonts. The participants correctly perceived that the second question was not distorted, as none of the participants reported that they were not able to answer it (see [Table 3](#)).

3.3. Equivalence testing

Failure to find support for the hypothesis does not mean that we found support against the existence of the hypothesized effect. Therefore, we performed equivalence testing using two one-sided tests (TOST, [Lakens et al., 2018](#)). TOSTs test whether the difference found in the replication study is smaller than the smallest effect of interest ([Lakens et al., 2018](#)). Two significant one-sided tests can be interpreted as evidence against the hypothesized effect. Such a result means that the tested effect is negligible.

To estimate the smallest effect of interest, we used a "small telescopes" approach ([Simonsohn, 2015](#)). According to the "small telescopes" approach, the smallest effect of interest can be determined as the effect that could be found in the original study with 33% test power. The replication study needs to have at least a 2.5 times larger sample than the original study to test for equivalence with 80% test power ([Simonsohn, 2015](#)). The sample size of the original study was $N = 32$. Therefore, our sample of 87 participants was sufficient.

In line with the small telescopes approach, we determined the smallest effect of interest using parameters from the original study. The participants in the original study were divided in two groups of 17 and 15 (the easy-to-read and difficult-to-read fonts respectively). Six of the 15 participants in the latter group identified the error (i.e., $p_2 = 0.4$) ([Song & Schwarz, 2008](#)). According to our analysis (G*Power 3.1.9.7, [Faul et al., 2007](#)), the proportion difference that could be detected in the original study with 33% test power was 24.6% (i.e., $p_1 = 0.154$, $p_2 - p_1 = 0.246$).

We used [Jamovi \(2020\)](#) to perform two one-sided tests of the difference of two proportions and tested whether the difference in detecting the Moses illusion between participants in both experimental conditions in our replication was less than 24.6 percentage points. Both one-sided tests of the difference of two proportions were significant at the 5% level (i.e., the difference was significantly smaller than $\pm 24.6\%$; upper: $z = 2.91$, $p = .002$; lower: $z = -1.78$, $p = .038$), which indicates support against a relevant effect of font readability on detection of error in a text.

Table 1
Effect of mood on error detection in the difficult-to-read condition.

	B	SE	Wald	df	Sig.	Exp(B)	95% CI Exp(B)	
							Lower bound	Upper bound
Mood	0.028	0.256	0.012	1	0.914	1.028	0.623	1.697
Constant	-0.471	1.569	0.090	1	0.764	0.625		

Table 2

Effect of mood on error detection in the easy-to-read condition.

	B	SE	Wald	df	Sig.	Exp(B)	95% CI Exp(B)	
							Lower bound	Upper bound
Mood	−0.276	0.291	0.905	1	0.341	0.758	0.429	1.340
Constant	1.053	1.760	0.358	1	0.550	2.865		

Table 3

Frequency table.

	Moses question		
	Number	I don't know	I can't answer
Easy-to-read font	54% (23/43)	9% (4/43)	37% (16/43)
Difficult-to-read font	46% (20/44)	11% (5/44)	43% (19/44)
	Switzerland question		
	Switzerland	I don't know	Other country
Easy-to-read font	93% (40/43)	7% (3/43)	0% (0/43)
Difficult-to-read font	77% (34/44)	14% (6/44)	9% (4/44)

4. Discussion

The goal of our study was to replicate the effects from the original study by [Song and Schwarz \(2008\)](#) of font-based Moses illusion with a larger sample and contribute to testing the effect of mood on detecting error in a text. In accordance with the original study ([Song & Schwarz, 2008](#)), we assumed that it would be easier to detect the error when the text was written in a difficult-to-read font, when compared to a text written in an easy-to-read font. However, unlike the original study, we did not find support for this hypothesis. Font readability did not significantly affect error detection.

Research on the Moses illusion has not found a unanimous explanation for this error in cognitive reasoning. According to some research, there is a greater chance of detecting a possible error in the text when the text is visually ordinary (black, regular font) or easy-to-read ([Haque et al., 2018](#); [Kanfer & Ackerman, 1989](#); [Surber, 1992](#)). An atypical font or a distinctive font colour does not increase the chances of error detection ([Eslick et al., 2011](#); [Gasser, Boeke, Haffernan, & Tan, 2005](#); [Price et al., 2015](#)). However, according to some other studies the chances of error detection increase with a difficult-to-read font ([Claypool et al., 2004](#); [Garcia-Marques & Mackie, 2001](#); [McGlone & Tofghbaksh, 2000](#); [Whittlesea et al., 1990](#)). Our results did not support the findings of those studies. On the contrary, our study suggests that the visual properties of a text are not essential for error detection.

One explanation for our results may be related to the manner in which we gave the instructions. According to [McCabe & Smith \(2002\)](#), a warning of a possible error is sufficient to support error detection. We communicated the instructions verbally at the beginning of the experiment; it is possible that the participants would not have detected the error had we given the instructions in writing. Nevertheless, we replicated the original study's instructions and did not have the same results. Thus, our findings suggest that the difference in font readability might not be a sufficient explanation for this error, as we did not find support for the findings of the original study by [Song and Schwarz \(2008\)](#).

Another explanation for our results may be found in the theory of [Reder and Kusbit \(1991\)](#). They claim that people tend to overlook errors in a text and adapt the environment to their mental representations. Therefore, we can assume that the participants may have automatically "corrected" the incorrect wording of the question and did not detect the error. Thus, error detection in the text is dependent on people's prior knowledge. If the required knowledge (in this case, awareness of the biblical figures Moses and Noah) is stored in memory, but is not often and actively used, it could be problematic to recall it from memory ([Fazio, Barber, Rajaram, Ornstein, & Marsh, 2013](#)). As the Czech Republic is considered an atheistic nation ([The Pew Research Centre's Forum on Religion & Public Life, 2002](#)), we assumed that Czech people are aware of biblical characters and the related stories, but do not use this knowledge actively, and therefore our results could have been influenced by this. The presumption of the familiarity of the Czech population with the biblical context is also supported by CERMAT. In 2015, it included a question on the name of the biblical character Judas Iscariot into the compulsory part of a Czech high-school graduation exam because such knowledge was considered to be a part of common knowledge of Czech students ([Nový Amos, 2015](#)). Although, the participants in our study had the option to report that they did not know the answer on the questionnaire, only 10% of the people in our replication responded in this way. Therefore, we believe that Czech students know the popular biblical stories and characters, and thus there is no reason to believe that lack of knowledge could have affected the results. However, the lack of active use of this knowledge may have played a role. Furthermore, [Reder & Kusbit \(1991\)](#) found that participants' ability to detect erroneous information did not increase with prior understating of the presented facts.

The results of this study might be in line with a partial matching explanation, which emphasizes semantic similarity as the reason why the Moses illusion works. According to [Park & Reder \(2004\)](#), one way to improve the detection of distortion is by emphasizing

features that distinguish the distorted term from the term it replaced. In that case, the difficult-to-read font might increase the critical review of the question, but it does not influence the thematic overlap or the illusion.

Studies that consider the effect of mood on cognitive processing to be important emphasize that a sad mood, in particular, has a positive impact on attention and sensitivity to detail (Bless, Bohner, Schwarz, & Strack, 1990; Schwarz & Clore, 2007; Alter, Oppenheimer, Epley, & Norwick, 2006). Nevertheless, mood did not affect error detection in our study.

The results of this replication could have been affected by the order of the questions. As in the original study, we presented one undistorted question and one distorted question. It is possible that reordering the questions could have facilitated error detection, as the participants would have actively remembered the introductory instructions and the cautionary statement about the error. This could have led to a more critical review of the first question, which may have led to more detection of the error. Whether the order of the questions plays a role in detecting the Moses illusion is a topic for further research. However, the order of the questions cannot be the cause of the difference between our results and those of the original study because it was the same in both studies.

This study has several limitations. Firstly, it only focused on one font style that was supposed to lead to better error detection. As previously mentioned, the readers' familiarity with the font style might play a role in the detection of mistakes. Future studies should attempt to verify the findings of our study with different font styles.

Secondly, participants were presented with only two questions (one of which was distorted). Therefore, the probability of detecting an error was more limited than it would be for a larger number of questions. Apart from the lack of question diversity, the questions in this study were not normed for the population, which might therefore have influenced its precision and resulted in insufficient grounds for generalization. However, the original study by Song and Schwarz (2008) also only presented two questions that were not normed for their population. As we were trying to replicate the original study in the Czech environment, we believe our study was not impaired by this limitation.

Thirdly, our results could have been influenced by the warning of possible error, which was given only at the beginning of the experiment and thus might not have drawn sufficient attention to the error.

A final limitation relates to how people rely on specific knowledge to detect error. Our research assumed that participants would have a basic knowledge of common biblical stories. However, as mentioned above, passive information is more difficult to recall, and in the Czech Republic, which is one of the least religious countries in Europe, it may be assumed that biblical knowledge is poorly stored in memory and not often recalled.

5. Conclusion

Our results did not support the hypothesis that an unusual font would attract the attention of readers and make it easier for them to detect a factual error. It seems that the more essential aspects of error detection might be actively used knowledge, perceived font familiarity and the warning about possible errors prior to the main task. Therefore, future research should focus more on clarifying the role of long-term memory and perception (the visual processing of information) in error detection.

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CRedit authorship contribution statement

Adela Janouskova: Conceptualization, Methodology, Writing – review & editing. **Jakub Kocyan:** Conceptualization, Methodology, Writing – review & editing. **Magdalena Simova:** Conceptualization, Methodology, Writing – review & editing. **Kamila Uvirova:** Conceptualization, Methodology, Writing – review & editing, Project administration, Funding acquisition. **Kamila Zahradnickova:** Conceptualization, Methodology, Writing – original draft. **Martin Vaculik:** Writing – review & editing, Funding acquisition, Supervision. **Jakub Prochazka:** Writing – review & editing, Software, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Questionnaire**Front side**

Field of study: _____

Age: _____

Sex: _____

Mood: 1 2 3 4 5 6 7 8 9

Very bad

Very good

Back side

Group 1: Allura font

Group 2: Arial font

*1. Která země je slavná pro své kukaččí hodiny, čokoládu, banky a kapesní nožičky?**2. Kolik zvířat od každého druhu vzal Mojžíš na archu?*

1. Která země je slavná pro své kukaččí hodiny, čokoládu, banky a kapesní nožičky?

2. Kolik zvířat od každého druhu vzal Mojžíš na archu?

Appendix B. Pilot testing

Group 1: Allura font

Švýcarsko je známé díky kukačkovým hodinám, bankám a kapesním nožikům.

Group 2: Arial font

Švýcarsko je známé díky kukačkovým hodinám, bankám a kapesním nožikům.

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